

Due on Tuesday, May 5th.

If marked, either questions 1+3+4 or questions 5+6 will be marked.

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1. Collision/decay problems.

- (a) [10 pts] Two photons have the same energy E_γ . They collide at an angle θ and create a single particle. Calculate the mass M of the final particle in terms of E_γ and θ .

- (b) [10 pts] Two balls of equal mass (m each) approach each other with equal but opposite velocities of magnitude v . Their collision is perfectly inelastic, so they stick together and form a single body of mass M . What is the velocity of the final body and what is its mass M ?

Find the mass M in the specific cases of $v = 0.01c$, $v = 0.5c$, and $v = 0.9c$.

2. The four-vector A^μ is found to be timelike in one inertial frame.

- (a) [4 pts] Write an inequality expressing the fact that A^μ is timelike.
- (b) [6 pts] Explain why A^μ has to be timelike in any inertial frame.

3. [SELF] The mass m and the charge q of a particle are four-scalars. Explain why the combination

$$(m, q, m, q)$$

is not a four-vector.

4. Electromagnetism.

- (a) [10 pts] Show that the tensor equation $\partial_\mu J^\mu = 0$ is equivalent to the continuity equation of electromagnetism. Here J^μ is the current density 4-vector.

- (b) [10 pts] The two inhomogeneous Maxwell's equations can be expressed as the tensor equation for the field tensor $F^{\mu\nu}$:

$$\partial_\mu F^{\mu\alpha} = J^\alpha$$

Derive the continuity equation from this equation, using the fact that the field tensor is antisymmetric.

Note: Depending on the definition conventions for $F^{\mu\nu}$ and J^μ , Maxwell's inhomogeneous equations may also appear in textbooks in the forms

$$\partial_\mu F^{\mu\alpha} = -J^\alpha \quad \text{or} \quad \partial_\mu F^{\mu\alpha} = \mu_0 J^\alpha \quad \text{or} \quad \partial_\mu F^{\mu\alpha} = -\mu_0 J^\alpha$$

but this will not matter for the problem above.

5. Minkowski tensors.

- (a) [12 pts] Consider the Minkowski tensors

$$D^{\mu\nu}, \quad D^{\mu\nu} B_\sigma, \quad D^{\mu\sigma} B_\sigma, \quad D^{\mu\sigma} D_{\mu\sigma}, \quad C^\mu B_\mu, \quad C^\mu B_\sigma.$$

Explain the rank of each tensor.

Which of these, if any, are scalars?

- (b) [6 pts] Show how the tensor $T_\gamma^{\alpha\beta}$ transforms under the Lorentz transformation Λ .
- (c) [8 pts] A Minkowski tensor $U_{\mu\nu}$ is said to be symmetric if $U_{\mu\nu} = U_{\nu\mu}$. If a tensor is symmetric in one inertial frame, show that it is symmetric in all inertial frames.

6. Poincaré transformations.

If Λ is a 4×4 matrix representing a Lorentz transformation, then transformations of the type

$$x' = \Lambda x + a$$

are known as Poincaré transformations. Here x and x' are 4×1 column vectors (4-vectors) representing spacetime coordinates of events as seen from two-different frames, and a is a 4×1 column vector. In other words, a Poincaré transformation is a combination of a Lorentz transformation plus a possible shift of the space and time coordinates. We will denote this transformation as (Λ, a) .

- (a) [10 pts] Show that the result of two Poincaré transformations (Λ_1, a_1) and (Λ_2, a_2) , applied successively, is the Poincaré transformation

$$(\Lambda_2 \Lambda_1, \Lambda_2 a_1 + a_2)$$

- (b) [6 pts] Are Poincaré transformations commutative?
- (c) [8 pts] Are Poincaré transformations associative?
- (d) [SELF] Does the set of all Poincaré transformations form a group?